Hriduan Omor Siam



Electrical & Electronic Engineering (Final Year Student) | VLSI & Design Verification
American International university-Bangladesh, Dhaka, Bangladesh
Email: hriduansiam101@gmail.com | Phone: +880 140 072 9346
LinkedIn: LinkedIn Profile
Website: E-Portfolio Website

ABOUT ME

I am a final-year EEE student at AIUB with a CGPA of 3.87. I have become really interested in VLSI and digital design. For my FSMD controller project I wrote the design in SystemVerilog and then used Cadence tools to synthesize it and simulate the timing paths. I also led my group at VLSITHON 3.0. I like tracking down why a design is not behaving the way it should and I want to build my career in the VLSI industry.

EDUCATION

American International University-Bangladesh (AIUB) | Dhaka, Bangladesh
BSc in Electrical and Electronic Engineering (EEE) — Final Year (CGPA: 3.87)

Mymensingh Polytechnic Institute | Mymensingh, Bangladesh
Diploma in Electronics Engineering — Graduated in 2023 (CGPA: 3.18)

Saint Andrews's High School | Haluaghat, Mymensingh
Secondary School Certificate (Science) — Graduated in 2017 (GPA: 4.55)

PROJECTS

- **FSMD Controller for ASIC & FPGA:** Built a Finite State Machine with Datapath (FSMD) controller that targets both ASIC and FPGA. I wrote it in SystemVerilog and then used Cadence tools to synthesize it and simulate the timing paths.
- **Digital Voting Machine:** Made an electronic voting machine that scans a voter's NID card and verifies them automatically. My goal was to make voting more secure and reliable.
- **Audio Signal Filtering:** Designed bandpass and low-pass filters in MATLAB to clean up and filter audio signals.
- **Smart Agriculture Model:** Put together a small smart farm prototype with automatic irrigation and a food dispenser. I also added an RFID card system to control who can enter the farm.
- **Ultrasound Sonar & RFID Access:** Made an ultrasound sonar system with Arduino and connected it to an RFID access control setup.

- **Solar Street Light & Phone Charger:** Built a solar-powered street light from basic components like diodes and op-amps. It also works as a phone charging station.

LEADERSHIP & EXPERIENCES

- **VLSITHON 3.0 Group Leader:** Took part in VLSITHON 3.0 organized by Ulkasemi and led my group.
- **Semiconductor Symposium Attendee:** Attended the National Semiconductor Symposium & BEAR Summit 2026.
- **Poster Presentation Participant:** Presented a poster as an author at the Bangladesh Youth Nuclear Congress (BYNC) competition under I4N x INNOVENTURE Bangladesh 2026.
- **Continuous Learner:** I go to engineering seminars at my university to learn about industry trends and topics we do not cover in class.
- **Student Representative:** In the final year of my diploma my teachers chose me to look after the daily running of our hostel. I handled day-to-day management and organized events for the students.
- **Industrial Tour:** Joined a university-organized industrial tour to PGCB (Power Grid Company of Bangladesh).

SKILLS

- **Programming & HDL:** SystemVerilog, C/C++ and MATLAB (all at intermediate level).
- **Digital Design:** Combinational and sequential logic, FSM and datapath design and RTL design for ASIC and FPGA.
- **EDA & Simulation Tools:** Cadence Xcelium for simulation, Genus for synthesis, Innovus for place and route, Virtuoso, AMD Vivado Design Suite, RealVNC, Multisim and Tinkercad.
- **Hardware Skills:** Circuit design, microcontrollers, RFID and solar energy setups.
- **Soft Skills:** Teamwork and leadership, clear communication, documentation and paperwork, event organizing.

DECLARATION

I declare that all the information above is true to the best of my knowledge.



Hriduan Omor Siam